Daryl BOULOU

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EXPERIENCE

EQUANS | *AI Agent Development* — **France** | *Sept. 2025 – Jan. 2026*

- **AI Strategy & Developer Enablement**: Recruited to design an open-source AI agent using Ollama for CI/CD optimization, then adapted to internal governance requirements by leading technical workshops on advanced GitHub Copilot capabilities to streamline engineering workflows.
- **Prompt Engineering for CI/CD**: Designed structured, reusable prompt templates to assist engineering teams with automated software delivery and pipeline tasks.
- **Tech Stack**: GitHub Copilot, Ollama, GitHub Actions, Prompt Engineering, Git.

AQUILAB | *QA Automation Intern* — **France** | *Jan. 2025 – Apr. 2025*

- **Test Automation & Release Validation**: Built end-to-end automated test scripts with Cypress and executed functional test cycles to validate production software releases.
- **Full-Stack Feature Development & UI/UX**: Revamped the application interface for a smoother user experience and implemented a new product feature to expand platform capabilities.
- **Tech Stack**: Cypress, TypeScript, JavaScript, UI/UX Design, Automated QA Testing.

EDUCATION

CESI School of Engineering | *Engineering Degree in Computer Science — AI & Data Science* — **France** | *2024 – Present*

Politehnica University of Timişoara | *Academic Exchange Semester* — **Timişoara, Romania** | *Mar. 2026 – Jul. 2026*

- **Relevant Coursework**: Natural Language Processing, Reinforcement Learning, Cloud Computing.

Prépa Saint-Jean | *Preparatory Classes for Engineering Schools* — **Douala, Cameroon** | *2022 – 2024*

PROJECTS

- **CameroonianLLM : Hybrid CPT & SFT Fine-Tuning of a 12B LLM for Camfranglais** — Hugging Face | GitHub
  Adapted `Mistral-Nemo-Instruct-2407` to Camfranglais through a two-stage QLoRA and rsLoRA pipeline on an NVIDIA L4 GPU, combining Continued Pre-Training on 23,224 texts with Supervised Fine-Tuning on 2,404 dialogues. Built an automated data pipeline that collected 163,283 YouTube comments alongside pseudonymized WhatsApp exports and filtered them via a custom linguistic scoring formula weighting 480 Wiktionary markers. Reached an 81.27% Camfranglais marker rate closely matching human references across 267 test dialogues, and published the GGUF model and corpora on Hugging Face.
- **AI Chess Coach : Local LLM & Stockfish Analysis** — GitHub
  Developed a full-stack interactive chess analysis web application combining real-time Stockfish streaming with a locally hosted `gemma4:e4b` model via Ollama. Engineered a blind-first agentic tool-calling loop to ground every explanation in UCI engine evaluations and eliminate hallucinations, packaged into a single multi-stage Docker container with React 19, TypeScript, FastAPI, Stockfish, and Ollama.
- **Leyenda : Deep Learning Image Processing & Automated Captioning** — Live Demo | GitHub
  Built and deployed a three-stage Computer Vision and NLP pipeline combining a 5-class CNN classifier with ElasticNet regularization and Youden thresholding at 93.04% binary accuracy and 0.9758 ROC-AUC, a fully convolutional U-Net image denoiser trained with combined MSE and SSIM loss, and an InceptionV3 with Bahdanau Spatial Attention and LSTM decoder trained on MS-COCO achieving a 0.750 CIDEr score. Deployed via Flask, Docker, Azure Container Apps, and Cloudflare.
- **Local GitLab AI Coding Agent** — GitHub
  Engineered an on-premise autonomous coding agent in Python using local LLMs and the `python-gitlab` API to monitor `ai:agent` issues, inspect repository trees, commit code fixes on dedicated branches, and automatically open Merge Requests.
- **MLP Language Model from Scratch** — GitHub | Article
  Implemented the Bengio et al. 2003 character-level Multi-Layer Perceptron language model from scratch in PyTorch using raw tensor operations, Kaiming initialization, custom Batch Normalization with exponential moving statistics, and Softmax logit scaling.

TECHNICAL SKILLS

- **Languages**: Python, TypeScript, JavaScript, SQL, Bash, HTML/CSS
- **AI & Machine Learning**: PyTorch, Hugging Face, QLoRA & PEFT, Keras 3, TensorFlow, Ollama, Computer Vision, NLP, Reinforcement Learning
- **Frameworks & Testing**: FastAPI, Flask, React, Cypress
- **Cloud & DevOps**: Docker, Azure Container Apps, Cloudflare, GitHub Actions, GitLab CI/CD, Linux
- **Spoken Languages**: French as native language, English with a 935 TOEIC score